

Topside sounder retrieval

TOPSIDE SOUNDER RETRIEVAL | 25 SEPTEMBER 2026

Status, recovery evidence, and truth-independence audit

Executive summary

The retrieval code can generate and fit synthetic vertical and oblique topside ionograms. This report evaluates one reduced vertical case and asks whether a low ionogram mismatch also means that the electron-density profile was recovered. It does not establish Mars retrieval accuracy.

Principal findings

Closed-model recovery

A true density scale of 1.12 was recovered as 1.00 by a six-point grid search (absolute error 0.12).

Profile-shape mismatch

A separately specified altitude perturbation produced a 6.7% profile NRMSE over 150–450 km, despite a small ionogram objective.

No-return failure

An empty truth ionogram and an empty candidate scored zero objective while their profiles differed by 32.0% NRMSE.

Validation verdict

The current demo's truth is generated by the same density parameterization, PyIRI background, ray tracer, and image formation used in retrieval. The alternate profile test changes only the density formula. Independent empirical Mars truth has not been established.

Read pages 2–5 for setup, figure interpretation, and the validation gate.

1. Methods and provenance

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One generated 800 km Earth orbit supplies a vertical transmitter/receiver case. The background is PyIRI; PHaRLAP traces three frequencies (4, 5, and 6 MHz). Only density scale is searched, at 0.80, 0.90, 1.00, 1.10, 1.20, and 1.30. This is a grid-search diagnostic, not a run of the full five-parameter optimizer.

Where each truth comes from

Case	Density truth	PyIRI	PHaRLAP	Mars data
Closed model	Fitted scalar family	Shared	Shared	No
Shape mismatch	Separate altitude formula	Shared	Shared	No
No-return control	Separate altitude formula	Shared	Shared	No

Table 1. Truth provenance. “Shared” means the truth generator and candidate model use the same component.

Scoring and accuracy measures

The ionogram objective combines normalized total-power error, O/X split error, and Doppler error. The profile NRMSE is root-mean-square density error divided by root-mean-square truth density, evaluated at the center grid column from 150 to 450 km. The ionogram objective has no direct percent interpretation.

Independence conclusion

A separate formula is useful for testing shape mismatch, but both sides still share Earth geometry, PyIRI, PHaRLAP, and ionogram construction. It is not a genuinely independent Mars ground truth.

2. Closed-model recovery

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First, the truth density is generated by the same scalar family searched by the retrieval. The generating scale (1.12) is deliberately absent from the six candidates.

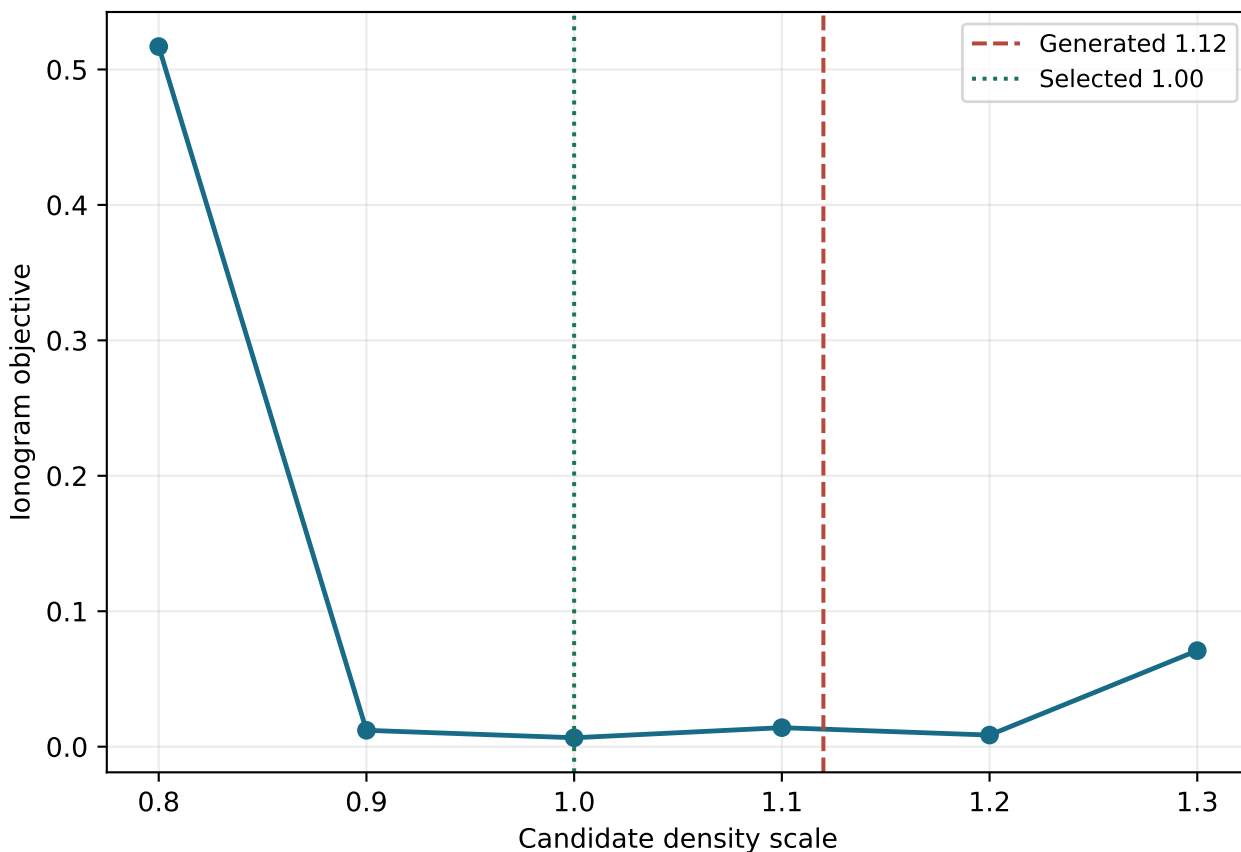


Figure 1. The six-point search selected scale 1.00 for truth 1.12. The objective is not smooth: neighboring candidates can change the available ray returns. This plot shows the complete set of evaluated candidates, not a continuous fit curve.

Interpretation

The absolute scale error is 0.12 (10.7% of truth). Because the generator and fitter share the same physical and numerical assumptions, this is a check of recoverability in one favorable model family, not independent validation.

3. Separate profile-shape test

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The truth profile is generated from the PyIRI background with an altitude-dependent factor, $1.10 + 0.10 \exp[-0.5((h - 320 \text{ km})/55 \text{ km})^2]$. Retrieval candidates can only multiply the background by a constant scale.

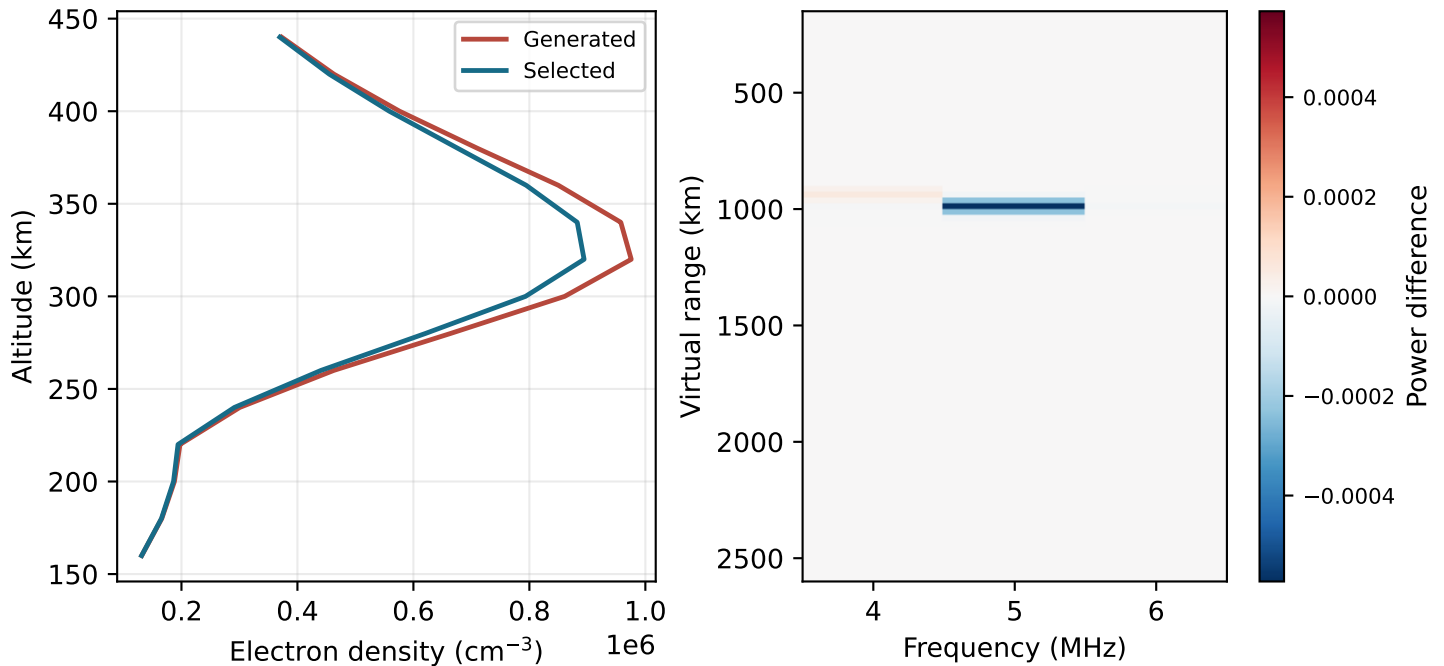


Figure 2. Left: the selected scale (1.10) misses the generated altitude-dependent profile by 6.7% NRMSE over 150–450 km. Right: fitted minus generated normalized ionogram power. Color is centered at zero; white regions have no visible difference at this scale.

Interpretation

The ionogram objective is 0.00014834, yet the profile error is 6.7%. A close observable fit does not establish the correct vertical density shape. This generator is independent only of the fitted scalar parameterization; it shares the background and ray tracer.

4. No-return failure and next gate

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A second altitude-dependent truth gives no returns at 4–6 MHz. The scale-0.80 candidate also gives no returns, so both ionograms are empty and their objective is zero.

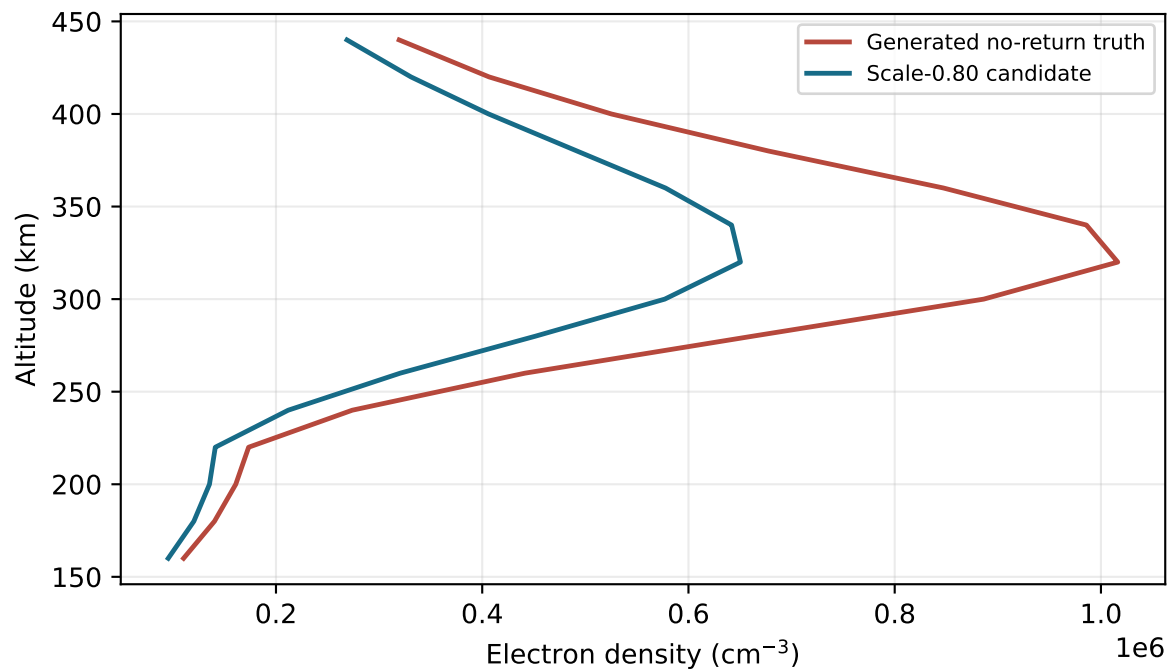


Figure 3. Profiles differ by 32.0% NRMSE from 150 to 450 km while the ionogram objective is exactly zero. The score cannot distinguish two empty observations. Such cases must be flagged and excluded from accuracy claims.

Conclusion

The present evidence supports a working synthetic diagnostic and exposes weak identifiability. It does not confirm independent Mars truth or measured retrieval accuracy.

Required validation

Use an external Martian density model or measured MARSIS profiles with Mars geometry. Hold the truth source outside the retrieval forward model, include frequencies with detected returns, and report density bias and NRMSE across multiple cases.

Public model lead

Huang et al. (2021), doi:10.1029/2021JA029226, publish dayside ion-density expressions for 150–450 km. Units, charge balance, and reuse terms need checking before use.